ECM3412

UNIVERSITY OF EXETER

FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

COMPUTER SCIENCE

Examination, May 2024

Nature Inspired Computation

Module Leader: Ayah Helal

Duration: TWO HOURS

Answer ALL questions.

The marks for this module are calculated from 60% of the percentage mark for this paper plus 40% of the percentage mark for associated coursework.

Candidates are permitted to use *non-programmable* portable electronic calculators in this examination.

This is a CLOSED BOOK examination.

Question 1

(a) What are the advantages of Nature-Inspired techniques? State an example of method and problem.

(5 marks)

(b) In Evolutionary algorithm, selection procedure is used to choose candidates for mutations and crossover. For years, the fitness proportionate selection (Roulette wheel selection) used to be the main method to be used. Can you explain this method, and state some disadvantages of this selection procedure?

(5 marks)

(c) Describe two applications of optimization algorithms in the context of machine learning, providing two examples for each application.

(6 marks)

(d) What is the difference between the fitness function of Genetic algorithm and Genetic programming ?

(4 marks)

(e) Explain the role of the **personal best** and **global best** particles. What is the effect of using both, and how is information from both combined to update particle position?

(4 marks)

(f) Assuming all objectives are to be minimised, explain how the *dominance* relation is used to compare solution quality in multi-objective problems.

(4 marks)

(g) With the aid of a diagram, explain the principle of *early stopping* and state why it is important.

(4 marks)

(h) Explain the process by which weights are updated in a self-organising map.

(4 marks)

(i) Explain how state transition rules in a cellular automata at time t are applied to obtain cell states at time $t + 1$.

(4 marks)

(Total 40 marks)

Question 2

(a) What are the properties of swarm intelligence algorithm family ? name two approaches?

(6 marks)

(b) In our lectures, we explored the use of Ant Colony Optimization (ACO) to optimize single machine scheduling. Expanding upon it, this question is about the design of ACO to address the scheduling of J jobs across M machines in a multiple machine job scheduling problem. Given the following J jobs and M machines:
2 Machines with 15 hours each starting at 9am.

Job	Time	Deadline
A	4 hours	3pm
B	3 hours	3:30pm
C	2 hours	5pm
D	2.5 hours	4pm
E	2 hours	4:30pm
F	3 hours	2pm
G	4 hours	1pm
H	3 hours	12pm

(i) How would you construct an ACO graph for this problem ? Mention any constrains the ants could have in the graph.

(6 marks)

(ii) Propose a fitness function to address task lateness within the context of the multiple machine job scheduling problem.

(3 marks)

(iii) Calculate the latency of each task, and calculate the fitness function for these two candidate solution.
M1 (ACDE)
M2 (BFGH)

(15 marks)

(Total 30 marks)

Question 3

(a) Particle Swarm Optimisation (PSO) uses the following formula to compute the velocity of a particle at the next timestep:

$$v_{ij}(t+1) = v_{ij} + c_1 \times z_1 \times (p_{ij} - x_{ij}(t)) + c_2 \times z_2 \times (g_{ij} - x_{ij}(t)). \quad (1)$$

For the particle

$$\mathbf{x}_i(t) = (0.3, 0.5, 0.7),$$

its associated velocity vector

$$\mathbf{v}_i(t) = (0.2, 0.5, 0.8),$$

the appropriate coefficients

c_1	0.9
c_2	0.1
z_1	0.35
z_2	0.65

,

and the *global best* and *personal best* solutions $\mathbf{g}$ and $\mathbf{p}_i$

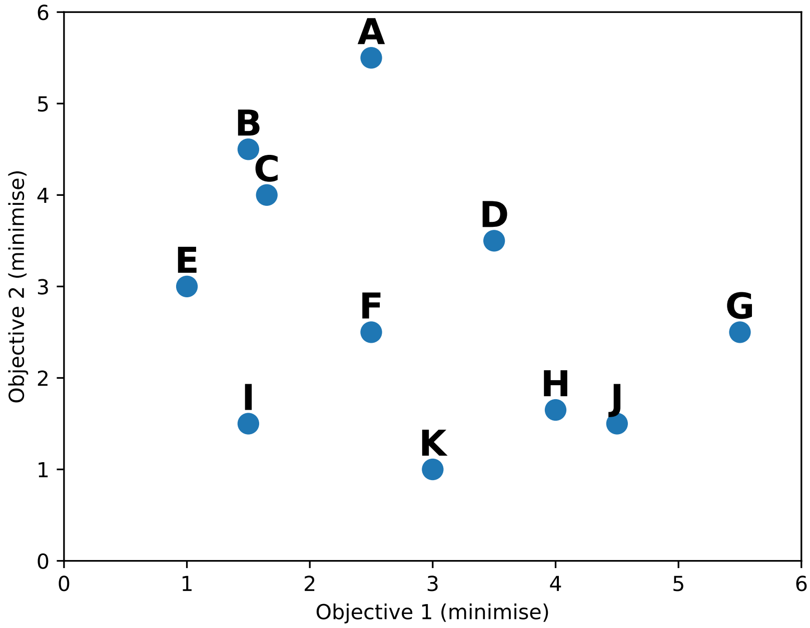
$$\mathbf{g} = (0.25, 0.55, 0.75)$$

$$\mathbf{p}_i = (0.25, 0.5, 0.85)$$

compute the new position of the particle at time $t + 1$.

(12 marks)

(b) The following search population is used by **NSGA-II** to optimise a two-objective problem. Both objectives are to be minimised.



Assuming NSGA-II is using a population size of 6, which solutions will be retained as parents in the next generation. Explain your answer.

(5 marks)

(c) Explain the principle of *graceful degradation* and propose an experiment to demonstrate that an artificial neural network has this characteristic.

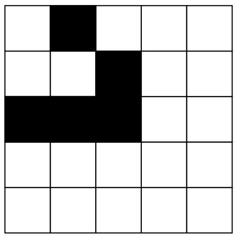
(7 marks)

(d) The initial state ($t = 0$) of an instance of *Conway's Game of Life* is shown. Using the Moore neighbourhood, and given the update rules:

- If a cell is off and exactly three of its neighbours are on then that cell becomes on in the next timestep – otherwise it remains off;
- If a cell is on and has either two or three active neighbours then it remains on – otherwise it becomes off.

Show the state if the CA for $t = 1$, $t = 2$, and $t = 3$.

$t = 0$



(6 marks)

(Total 30 marks)